AGENT BILLING & COMPLIANCE DASHBOARD

Power BI Portfolio Case Study | Fly Analytics

PROJECT	ROLE	DURATION	TOOLS	STATUS
Fly Analytics – Energy Company Billing Operations	IT Analyst	1–2 Weeks / Production	Power BI, DAX, Power Query, DAX Studio	Active – used monthly

EXECUTIVE SUMMARY

This case study documents a production Power BI billing analytics solution delivered as IT Analyst for an energy utility company. The project addressed two critical needs simultaneously: a full audit and correction of the entire 2025 billing dataset which contained significant financial inconsistencies and the design and deployment of a new dashboard enabling management to track each agent's billing productivity in monetary terms for the first time.

The solution provides granular visibility into agent-level transaction costs, salary vs. output comparisons, compliance rates, and target achievement across teams, transforming what was previously a manual, error-prone monthly process into a reliable, data-driven operational tool.

27 Agents Tracked	$1.47B+ Salary Pool	59.72% Avg Compliance	$877M+ Transaction Cost	1–2 Weeks Dev to Prod

BUSINESS CHALLENGE

The Problem

An energy utility company managed billing for a team of 27+ agents processing transactions monthly. Before this solution existed, the organization faced two critical and compounding problems:

- **Corrupted billing data:** The full 2025 billing records contained financial inconsistencies and calculation errors making it impossible to reliably assess agent output or cost efficiency.
- **Zero productivity visibility:** Management had no way to see in monetary terms what each agent was producing relative to their salary. No tool existed to compare individual agent salary against transaction output, track monthly productivity targets, or identify high vs. low performers.
- **Manual reporting:** Monthly billing reviews were manual, time-consuming, and unreliable built on data nobody could fully trust.

Solution Approach

Delivered a 3-view Power BI billing dashboard in 1–2 weeks that solved both problems simultaneously: first auditing and correcting the entire 2025 billing dataset, then building a production-ready analytics tool now used every monthly billing cycle.

- **Phase 1 – Data Audit & Correction:** Identified and resolved financial inconsistencies across the full 2025 billing dataset, cross-referencing transaction records, correcting cost calculations, and validating agent assignment data.
- **Phase 2 – Dashboard Development:** Built a 3-view Power BI solution (Agent Summary, Target Dashboard, Agent Detail) featuring salary vs. transaction cost comparisons, compliance tracking, monthly point achievement, and team-level breakdowns deployed to production within the same sprint.

TECHNICAL IMPLEMENTATION

TOOL	PURPOSE	KEY USAGE
Power BI Desktop	Visualization & Reporting	Multi-view dashboard, conditional formatting, drill-through
DAX	Calculations & Measures	Compliance %, salary vs. cost delta, point achievement levels
Power Query (M)	Data Transformation	ETL, billing record correction, column derivations
DAX Studio	Performance Optimization	Query tuning, measure validation, load time optimization

Data Model Architecture

- Billing Records (full 2025, audited and corrected): Agent ID, Period, Transaction Cost, Unit Points, Classification (Agile / Non-Agile), Process Type, Channel (On-site / Virtual)
- Targets: Agent ID, Monthly Goal, Team Assignment (Team 1 / Team 2)
- Agents dimension: Agent ID, Full Name, Team
- Process Classification dimension: Classification ID, Name, Category
- Time dimension: Period, Month, Year

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Agent: All

Month: All

54,21%
(-22,55%)
Avg Compliance

Agent Summary Dashboard

Target Dashboard

Agent Detail Dashboard

Agent Summary Dashboard

Agents	Total Salary	Total Transaction Cost	Total Difference	Average Compliance
Lucy Carson	54.428.292	40.957.809,00	13.470.480,42	75,25 %
Paula Andrews	54.428.292	39.561.070,00	14.867.222,96	72,68 %
Ingrid Palmer	54.428.292	38.716.672,00	15.711.619,06	71,13 %
Edgar Walsh	54.428.292	38.082.417,00	16.345.875,83	69,97 %
Diana Foster	54.428.292	37.994.154,00	16.434.137,30	69,81 %
Frank Allen	54.428.292	37.303.101,00	17.125.190,32	68,54 %
Katherine Johnson	54.428.292	37.057.839,00	17.370.452,84	68,09 %
Victor Adams	54.428.292	35.872.955,00	18.555.334,70	65,91 %
Leidy Castro	54.428.292	35.334.636,00	19.093.655,42	64,92 %
Michael Bennett	54.428.292	34.451.485,00	19.976.807,27	63,30 %
Henry Morgan	54.428.292	34.019.722,00	20.408.569,44	62,50 %
Sandra Peterson	54.428.292	34.014.117,00	20.414.174,92	62,49 %
Daniel Brooks	54.428.292	32.778.215,00	21.650.076,39	60,22 %
Jesse Morris	54.428.292	32.232.175,00	22.196.117,95	59,22 %
Maurice Rivers	54.428.292	31.894.359,00	22.533.933,22	58,60 %
Roger Spencer	54.428.292	31.825.260,00	22.603.031,67	58,47 %
Jennifer Diaz	54.428.292	31.664.084,00	22.764.207,23	58,18 %
Omar Gomez	54.428.292	31.507.951,00	22.920.341,70	57,89 %
John Alexander	54.428.292	31.221.319,00	23.206.971,69	57,36 %
Marisol Carter	54.428.292	30.815.960,00	23.612.332,41	56,62 %
Nelson Rivers	54.428.292	30.487.063,00	23.941.229,10	56,01 %
Nelly Garrison	54.428.292	28.798.731,00	25.629.560,95	52,91 %
Elizabeth Hayes	54.428.292	26.684.037,00	27.744.254,94	49,03 %
James Cooper	54.428.292	26.512.207,00	27.916.087,60	48,71 %
Jessica Turner	54.428.292	24.448.373,00	29.979.919,03	44,92 %
Adriana Fonseca	54.428.292	23.414.921,00	31.013.372,26	43,02 %
Emily Parker	54.428.292	20.041.602,00	34.386.690,16	36,82 %
Total	**1.469.563.884**	**877.692.234,00**	**591.871.646,80**	**59,72 %**

Agent Summary Dashboard

The top-level view provides a full ranking of all 27 agents by billing efficiency. Each row shows Total Salary, Total Transaction Cost, Total Difference, and Average Compliance %. Top performer achieved 75.25% compliance, while the overall average sits at 54.21%, down 9.65% from the prior period, flagging a team-wide performance trend for management attention.

Fly Analytics

Agent: All

Month: All

Agent Summary Dashboard

Target Dashboard

Agent Detail Dashboard

Target Dashboard - Team 1

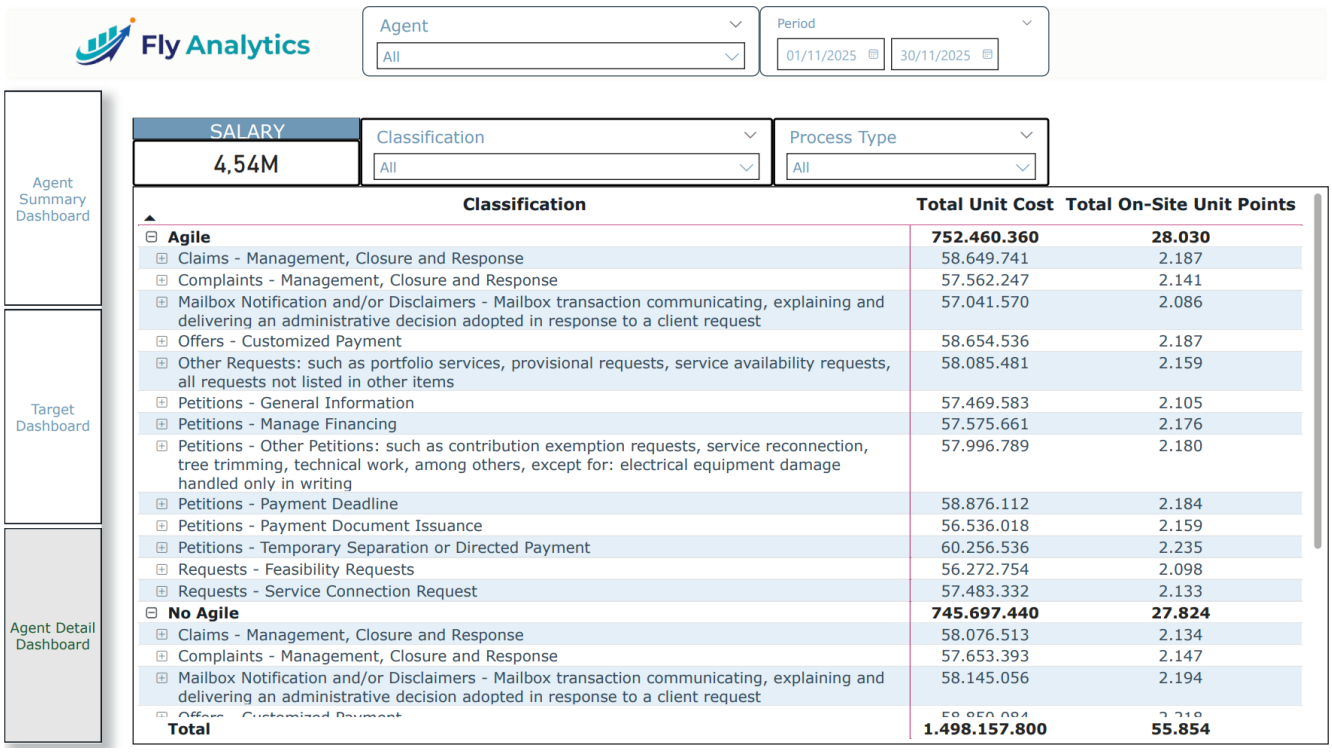
Agent	Goal	Total On-site	Total Virtual	Total Monthly Points	Achieved Level	Total Transactions	Point Differen
Sandra Peterson	54.516	36.486	8.860	45.346	9,98	45.712	-9.168,40
Daniel Brooks	54.507	35.044	9.655	44.699	9,84	44.996	-9.807,44
Paula Andrews	54.488	34.225	10.183	44.408	9,78	44.666	-10.079,80
Leidy Castro	54.509	37.604	5.694	43.298	9,53	43.589	-11.211,23
Nelly Garrison	54.512	33.996	8.661	42.657	9,39	43.021	-11.853,46
Edgar Walsh	54.484	33.585	8.045	41.630	9,17	42.000	-12.853,77
Elizabeth Hayes	54.519	32.367	8.866	41.233	9,07	41.495	-13.286,40
Lucy Carson	54.552	34.275	6.381	40.656	8,94	40.914	-13.894,80
James Cooper	54.506	30.933	9.550	40.483	8,91	40.877	-14.021,95
Michael Bennett	54.531	32.266	8.029	40.295	8,87	40.594	-14.234,94
Marisol Carter	54.497	29.497	10.072	39.569	8,71	39.876	-14.928,79
Nelson Rivers	54.528	29.768	7.313	37.081	8,16	37.374	-17.445,59
Jessica Turner	54.532	29.541	6.889	36.430	8,02	36.845	-18.102,14
Victor Adams	54.499	29.874	5.868	35.742	7,87	35.988	-18.755,88
Emily Parker	54.503	29.778	5.066	34.844	7,67	35.142	-19.658,59
Omar Gomez	54.509	27.043	7.166	34.209	7,53	34.620	-20.300,52
Frank Allen	54.515	27.269	6.164	33.433	7,36	33.697	-21.080,58
Adriana Fonseca	54.519	28.083	4.982	33.065	7,28	33.219	-21.453,21
Katherine Johnson	54.487	26.964	5.556	32.520	7,16	32.772	-21.966,79
Ingrid Palmer	54.507	25.196	6.611	31.807	7,00	32.099	-22.700,13
Diana Foster	54.488	24.731	6.999	31.730	6,99	32.043	-22.760,20
Jennifer Diaz	54.482	24.631	6.464	31.095	6,85	31.487	-23.387,49
Maurice Rivers	54.550	26.040	5.022	31.062	6,84	31.279	-23.488,59
Henry Morgan	54.499	22.770	6.641	29.411	6,48	29.751	-25.087,79
John Alexander	54.530	16.172	5.176	21.348	4,70	21.680	-33.181,96
Total	**1.471.798**	**811.658**	**200.330**	**1.011.988**	**222,78**	**1.020.292**	**-459.802,34**

Target Dashboard - Team 2

Agent	Goal	Total Monthly Points	Achieved Level	Total Transactions	Point Difference	Total On-Site	Total Virtual	Sum
Natalie Rose Henderson	13.560,00	10.210,47	9,04	10.206	-3.349,53	10.206	0	
Brianna Faith Coleman	54.428,29	38.684,15	8,53	38.678	-15.744,14	38.678	0	
Marcus Leon Bradley	13.560,00	8.993,76	7,96	8.987	-4.566,24	8.987	0	
Tyler James Woods	54.428,29	33.692,31	7,43	33.687	-20.735,98	33.687	0	
Sophie Claire Williams	13.560,00	7.037,21	6,23	7.032	-6.522,79	7.032	0	
Total	**149.536,58**	**98.617,92**	**39,18**	**98.590**	**-50.918,67**	**98.590**	**0**	

Target Dashboard – Teams 1 & 2

A dual-team view tracking monthly point achievement against individual goals. Team 1 (25 agents) totalled 1,011,988 monthly points with an average achieved level of 8.91 out of 10. Team 2 (5 agents) is tracked separately with a visible point deficit flagging underperformance for management review. On-site vs. virtual transaction splits are shown per agent.



Classification	Total Unit Cost	Total On-Site Unit Points
Agile	**752.460.360**	**28.030**
Claims - Management, Closure and Response	58.649.741	2.187
Complaints - Management, Closure and Response	57.562.247	2.141
Mailbox Notification and/or Disclaimers - Mailbox transaction communicating, explaining and delivering an administrative decision adopted in response to a client request	57.041.570	2.086
Offers - Customized Payment	58.654.536	2.187
Other Requests: such as portfolio services, provisional requests, service availability requests, all requests not listed in other items	58.085.481	2.159
Petitions - General Information	57.469.583	2.105
Petitions - Manage Financing	57.575.661	2.176
Petitions - Other Petitions: such as contribution exemption requests, service reconnection, tree trimming, technical work, among others, except for: electrical equipment damage handled only in writing	57.996.789	2.180
Petitions - Payment Deadline	58.876.112	2.184
Petitions - Payment Document Issuance	56.536.018	2.159
Petitions - Temporary Separation or Directed Payment	60.256.536	2.235
Requests - Feasibility Requests	56.272.754	2.098
Requests - Service Connection Request	57.483.332	2.133
No Agile	**745.697.440**	**27.824**
Claims - Management, Closure and Response	58.076.513	2.134
Complaints - Management, Closure and Response	57.653.393	2.147
Mailbox Notification and/or Disclaimers - Mailbox transaction communicating, explaining and delivering an administrative decision adopted in response to a client request	58.145.056	2.194
Offers - Customized Payment	58.850.084	2.218
Total	**1.498.157.800**	**55.854**

Agent Detail Dashboard

The most granular view drills into a single agent's monthly billing composition — showing unit cost and unit points broken down by process classification (Agile vs. Non-Agile) and specific service type such as Claims Management, Petitions, Service Connection Requests, and more. Enables precise cost-per-transaction analysis at the individual level.

BUSINESS IMPACT

IMPACT AREA	BEFORE	AFTER
Billing Data Quality	2025 records had errors & inconsistencies	Full year audited, corrected & validated
Agent Productivity Visibility	No monetary view of agent output	Real-time salary vs. transaction cost per agent
Monthly Reporting	Manual, time-consuming process	Dashboard updated each billing cycle
Performance Management	No compliance benchmark	54.21% avg tracked with trend alerts
Target Tracking	No goal measurement system	Point-based targets with team segmentation
Decision Making	Reactive, based on incomplete data	Proactive, data-driven management reviews

KEY TECHNICAL LEARNINGS

- **Data Quality First:** A clean, audited data foundation is a prerequisite for trustworthy analytics. Correcting the 2025 billing records before building the dashboard ensured that every metric displayed was reliable from day one.
- **Monetary KPIs Drive Action:** Translating agent performance into salary vs. transaction cost terms made the data immediately actionable for management — far more impactful than abstract productivity scores.
- **Segmented Views Scale Better:** Splitting the target dashboard by team and providing an individual agent drill-down meant the same tool served both executive oversight and individual performance coaching.
- **DAX for Dynamic Compliance:** Building compliance and point achievement as dynamic DAX measures — rather than static calculated columns — allowed the dashboard to stay accurate as new monthly billing data was loaded.

DATA PRIVACY NOTICE

All agent names, financial figures, and operational data presented in this case study have been anonymized and modified to protect client confidentiality. The dashboard architecture, methodology, DAX logic, and analytical approach reflect the actual production solution delivered.